Gamma Hedger

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1 Introduction

When it comes to trade options, you may often want to delta-hedge your position. It can be either used to lock gamma-pnl or instead to reduce as much as possible the market sensibility and to receive theta-pnl. In this report, we are not discussing how to generate profit with delta-hedging options but how to properly hedge an option at the close.

While it can be easy to dynamically hedge positions during trading hours, a challenge appears at the close and more particularly during the fixing time range: **Closing Auction**.
As a reminder, it represents the ten last minutes (3:50 pm - 4:00 pm) during which each market actors can send orders as "Market-On-Close" (MOC) and "Limit-On-Close" (LOC). As a result, a balance appears and the closing price will be the price which maximize the traded volume based on the accumulation of these types of orders.

1.1 Why does the Closing Auction matter?

Often, during Closing Auction, prices vary a lot due to high volume coming from hedge-fund and asset managers. Once the market is closed, your order cannot be executed for the day's session.

For deep out-the-money or deep in-the-money options, it doesn't matter. However, for near at-the-money options, it can be quite challenging because you cannot know in advance at which price the underlying will close and will have huge impact on your options.

1.2 Example

Suppose your want to delta-hedge at the end of the day a **long ATM Call** close to maturity. This kind of option is massively sensitive to the underlying. You do not want to (and cannot) send unlimited orders and recompute your delta continuously to match the exact price at the close and be delta-neutral. But at the same time, you absolutely need to be delta-neutral otherwise you are exposed to a maximal gamma risk.

Assume you are delta-neutral ten minutes before the close.

And:

$$\Gamma_{call} = 0.05$$

$$\delta_{call} = 0.5$$

$$\delta_{portf} = 0$$

It means that 1% change in the underlying will impact δ_{call} by 10%.

So, if at the terms of the Closing Auction, the price has risen by 0.5%, then:

$$\delta_{call} = 0.525$$

$$\delta_{portf} = 0.025$$

I am no longer delta-hedged (250 bps is huge)!

More interesting, even if I anticipated the 0.5% rise and sold more stocks, I would have not been hedged due to the high gamma of the portfolio.

So, here are the two main challenges:

- How to use the closing auction to be delta neutral outside RTH (Regular Trading Hours) no matter the closing price?
- How to take into account the gamma effect on my hedge?

2 Delta and Gamma of a Call

For the following parts, we will consider a long position on a call with its delta hedge. In other words, we will be long gamma.

By deriving the Black & Scholes formula, we mathematically get:

$$\delta_{call}(S_t, t) = \mathcal{N}(d_1(S_t, t))$$

$$\Gamma_{call}(S_t, t) = \frac{\phi(d_1(S_t, t))}{S_t \sigma \sqrt{T-t}}$$

where

$$d_1(S_t, t) = \frac{\ln\left(\frac{S_t}{K}\right) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$$

$$\phi : \text{ density of the standard normal distribution}$$

$$\mathcal{N} : \text{ density of the standard normal distribution}$$

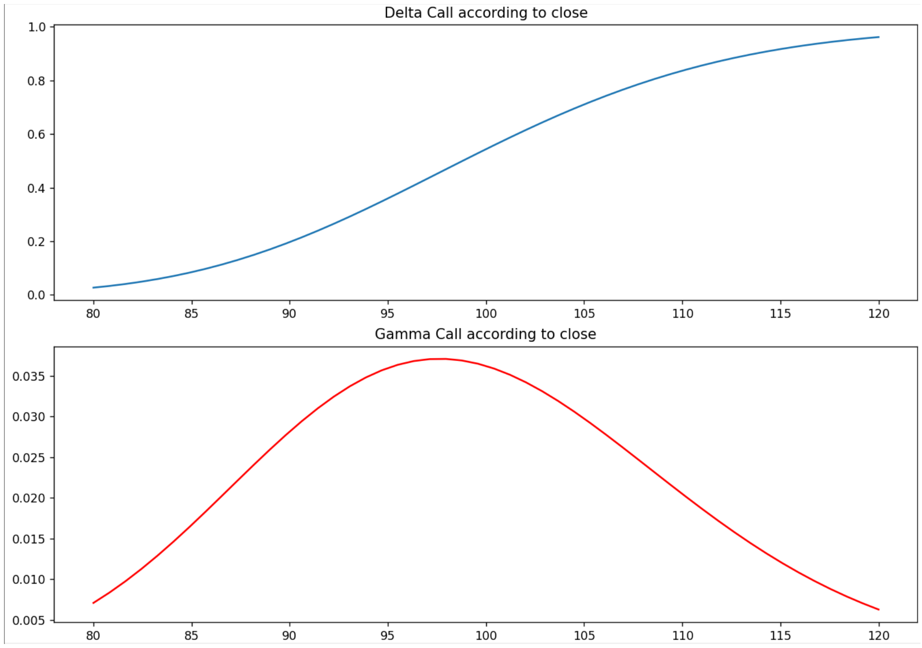

Figure 1: Delta and Gamma of a Call, $K = 100, r = 2\%, vol = 20\%, T - t = 0.3$

We can see it is essential to take gamma effect into account to hedge an option, particularly when the option is near at-the-money.

3 Gamma Hedger

I developed the tool in Python but we are not focusing on the implementation but rather on the results.

3.1 Context

Let's set the context of its use. We assume we are holding a portfolio of a **long contract call of size 100 on stock XYZ**.

The current data are:

- $K = \$100$
- $r = 3\%$
- Implied vol $= 60\%$

- $T - t = 0.01$ (3 days)
- $Div = 0\%$
- Spot XYZ = \$97

Assume now we only have a short position on stock of 30 and we are approaching the Closing Auction.

Dynamic Portfolio:

- 1 Call (size 100) on XYZ
- -30 stocks XYZ

Using Black-Scholes formula, its follows that:

$$\delta_{Portfolio} = 1.82 \text{ (number of shares)}$$

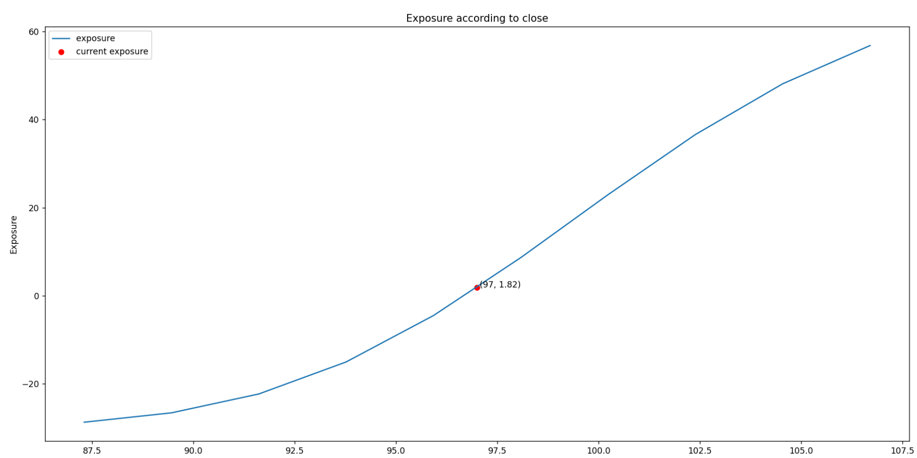

Figure 2: Profile of our portfolio's current exposure

We can see that there is no chance we fail our hedging at the close otherwise we are going to get huge market exposure if the underlying slightly moves.

So, now that we are ahead of the Closing Auction and we know how our delta exposure will evolve, we have to send a set of limit orders taking into account it.

The more the spot will increase, the more we will have to sell. Also, the more the spot will decrease, the more we will have to buy.

3.2 How to set (quantity, limit price) on each order?

Principles: Cut the portfolio's exposure into small segments alongside the potential closing price and compute the amplitude of each segment.

Each segment will correspond to an order:

- quantity = amplitude of exposure of the segment
- limit price = closing price of the external bond

Illustration for selling orders (increasing scenario):

We construct subset every 1% change of the closing price, starting from the breakeven close that would make the portfolio delta-neutral. As we started with a positive $\delta = 1.82$, the breakeven is \$96.70, lower that the current spot \$97.

Close (\$)	96.70	97.77	98.85	99.92
Exposure (nb. shares)	0	6.68	13.72	20.88

Table 1: Exposure of portfolio alongside closing price

<u>1st limit order to send: **S 7 @97.77**</u>

- quantity = rounded amplitude of exposure of first segment = $6.68 - 0 \to 7$
- price = external bond $\to$ \$97.77

<u>2nd limit order to send: **S 7 @98.85**</u>

- quantity = rounded amplitude of exposure of first segment = $13.72 - 6.68 \to 7$
- price = external bond $\to$ \$98.85

Illustration for buying orders (decreasing scenario):

Close (\$)	96.70	95.62	94.55	93.47
Exposure (nb. shares)	0	-6.13	-11.57	-16.20

Table 2: Exposure of portfolio alongside closing price

<u>1st limit order to send: **B 6 @95.62**</u>

- quantity = rounded amplitude of exposure of first segment = $|-6.13-0| \to 6$

- price = external bond → \$95.62

<u>2nd limit order to send: **B 5 @94.55**</u>

- quantity = rounded amplitude of exposure of first segment = $|-11.57-(-6.13)| \rightarrow 5$
- price = external bond → \$94.55

Limit Price (\$)	96.70	95.62	94.55	93.47	92.40	91.32	90.25	89.17	88.10	87.02
Quantity	0	6	5	5	4	3	2	2	1	1

Table 3: Buying Orders

Limit Price (\$)	96.70	97.77	98.85	99.92	100.99	102.07	103.14	104.22	105.29	106.36
Quantity	0	7	7	7	7	7	6	6	5	4

Table 4: Selling Orders

Overall results Assume the Closing Auction results with a closing price on XYZ of \$99.

Then, we are in the increasing scenario and only 2 selling orders would have been executed.

- **S 7 @97.77**
- **S 7 @98.85**

At the close, our portfolio would have been composed of:

- 1 Call (size 100) on XYZ
- -44 stocks XYZ

With

$$\delta_{Portfolio} = 0.73 \text{ (number of shares)}$$

3.3 Impact of the discretization step

While we well took into account the gamma effect for hedging, it is not perfect. The inaccuracy comes from our subsets every 1%. The lower the discretization step, the more accurate.
$h = 0.2\%$ seems to be an effective discretization step.

To illustrate it, let's take $h = 0.2\%$. We assume the closing price is \$99.10 to make it more concrete (rounded is applied to take decimals into account).

Limit Price ($)	96.70	96.89	97.09	97.29	97.48	97.68	97.88
Quantity	0	1	1	2	1	1	1
Limit Price ($)	98.08	98.27	98.47	98.67	98.87	99.06	99.26
Quantity	2	1	1	2	1	1	1

Table 5: Selling Orders

In this case, a total of 15 stocks would have been sold (12 limit orders with quantity of 1 or 2 executed).

At the close, our portfolio would have been composed of:

- 1 Call (size 100) on XYZ
- -45 stocks XYZ

With

$$\delta_{Portfolio} = 0.39 \text{ (number of shares)}$$

There is a reduction of exposure by 46%! But you are making more trades, costing you more fees. That is why there is always a choice to be made when delta-hedging: do I hedge or do I wait until it goes back or further?

Remarks

A gamma hedger is only useful when your portfolio is long gamma, **ie** selling when prices go higher and buying when prices go lower in order to adjust the delta. It works because the higher the closing price, the more selling orders will be executed. Similarly, the lower the closing price, the more buying orders will be executed.

The opposite does not work!